Supporting Information

Amplitude and midpoint statistics of the same chemotaxis cells demand incompatible receptor models

Huan Li

Shanghai Liangta Biotech Co., Ltd., Shanghai 201899, P. R. China

October 2026

This Supporting Information documents the complete computational output of the accompanying repository (`chemotaxis-receptor-layer`, release v1.0.0): every analysis script, the numerical results each produced, and the provenance of the third-party FRET archive. Numbers are transcribed verbatim from `results/`; scripts are plain Python 3 (numpy/scipy/matplotlib) and read the `.mat` files in `data_external/moore2024_fret/` directly.

Contents

1 Pipeline map: scripts and outputs **2**

2 P7 adjudication: $K_{1/2}$ medians vs the Weber line **2**

3 Adaptation-relay baseline audit **3**

4 Prospective mixed-ligand arms (code70) **3**

5 Third-party data provenance **3**

1 Pipeline map: scripts and outputs

Script	Computes	Archived output
`code49_P7_blind_adjudication_FCD_Weber`	P7 blind adjudication of the fold-change/Weber claim on the Moore 2024 archive	adjudication tables
`code49b_P7_erratum_readjudication`	erratum re-adjudication (censoring-aware)	the code49b unit-table CSV (Chinese-named, `results/P7_Moore2024_FCD-Weber/`)
`code51chemo_receptor_layer_selfcheck`	receptor-layer internal-consistency self-check	console log
`code52_protocol_artifact_test_raw_FRET_coordinate_analysis`	protocol-artifact test in raw FRET coordinates	console/tables
`code53_CheA_node_mechanism_kernel_competition`	CheA-node mechanism: kernel competition	console/tables
`code54_K12_joint_fit_rupture_quantification`	joint-fit rupture quantification on the $K_{1/2}$ ladder	console/tables
`code55_multispecies_local_adaptation_joint_test`	multi-species / local-adaptation joint test	console/tables
`code56_FRET_readout_nonlinearity_test`	FRET readout-nonlinearity control	console/tables
`code57_TCS_depletion_robustness_gain_retuning`	TCS-depletion robustness with gain retuning	console/tables
`code58_P8_independent_arm_serine_Lasp`	P8 independent arm (serine, L-aspartate)	console/tables
`code59_residual_waterfall_window_structure_rhoF`	residual waterfall; window structure of ρ_F	console/tables
`code60_heterogeneous_population_censoring_test`	heterogeneous-population censoring control	console/tables
`code61_rowwise_Hill_steepness_test`	row-wise Hill-steepness test	console/tables
`code62_single_cell_same_source_statistics`	same-source single-cell statistics	console/tables
`code70_mixed_ligand_validation_arm_P1P2`	prospective mixed-ligand validation arms P1/P2	`chemotaxis_prospective_arm.csv`
`chemotaxis_adaptation_audit.py`	adaptation-relay baseline audit on the full archive	`chemotaxis_adaptation_audit.json`

2 P7 adjudication: $K_{1/2}$ medians vs the Weber line

Censoring-aware re-adjudication (code49b) of the FCD–Weber claim on the Moore et al. (2024) single-cell FRET archive. Table 2 is the archived unit table (verbatim rows).

The per-cell unit table (the unit-table CSV under `results/P7_Moore2024_FCD-Weber/` (Chinese-named)) carries, per (background B, foreground F) condition: fold change r, threshold T, $\log_{10} r$, $\log_{10} T$, n cells,

Table 2: $K_{1/2}$ medians per background level vs Weber-line prediction.

Background (μM)	$K_{1/2}$ median (μM)	IQR lo–hi	censored / n	Weber pred. (μM)	dev. (dex)
0.0	2.033	1.583–3.105	179/268	2.282	−0.050
0.01	2.896	2.080–3.065	69/101	2.293	+0.101
0.1	2.168	1.769–2.844	35/57	2.399	−0.044

and the two arm statistics R_a, R_A; e.g. B=0.01, F=0.2 μM: $r = 21.0$, $T = 0.21$, $n = 101$, $R_a = 0.0655$, $R_A = 0.0209$.

3 Adaptation-relay baseline audit

Full-archive audit (`chemotaxis_adaptation_audit.json`): 2217 cells, 77,595 response events; median baseline activity after high- vs low-amplitude preceding stimuli 0.434 vs 0.448; median absolute response 0.608. The per-(preceding-dose) baseline bands are archived in the same JSON.

4 Prospective mixed-ligand arms (code70)

Frozen-then-measured prospective arm V1 (clause P-1b, amplitude table of the L-aspartate component). Verbatim rows of `chemotaxis_prospective_arm.csv`:

Table 3: Prospective arm V1: measured vs predicted amplitudes (tolerance: max |dev| = 0.08).

Foreground (μM)	measured	predicted	deviation	hit
5	0.047	0.119	−0.072	Y
10	0.085	0.152	−0.067	Y
20	0.139	0.187	−0.048	Y

5 Third-party data provenance

`data_external/moore2024_fret/` mirrors, verbatim, the 48-file single-cell FRET archive of Moore, Kamino, Kottou, Shimizu & Emonet, *Cell Syst.* 15, 628–638 (2024), Dryad doi:10.5061/dryad.nvx0k6dzz, under its CC0 1.0 public-domain dedication; please cite the original study. Receptor-layer parameter priors derive from Sourjik & Berg, *Proc. Natl Acad. Sci. USA* 99, 123–127 (2002); Shimizu, Tu & Berg, *Mol. Syst. Biol.* 6, 382 (2010); and Tu, Shimizu & Berg, *Proc. Natl Acad. Sci. USA* 105, 14855–14860 (2008); fold-change readout evidence from Lazova et al., *Proc. Natl Acad. Sci. USA* 108, 13870–13875 (2011). Full notes in `data_external/PROVENANCE.md`.